

Decision Memo: Drone Depot Swap Capacity

To: Operations Manager, Bosphorus Drone Light Festival
From: IE 306 Analytics Team
Date: May 30, 2026
Re: Whether to purchase one additional battery-swap station

Recommendation

Purchase the sixth swap station, provided its capital cost is reasonable relative to the value of reduced depot delay. The simulation evidence shows a clear and practically meaningful reduction in the steady-state swap-queue wait when moving from the current five-station depot to six stations.

Evidence

I treated the depot as a steady-state system and deleted the initial warmup caused by the synchronized full-battery start. Welch's method suggested 7,200 seconds, MSER suggested 7,620 seconds, and I used 7,620 seconds downstream. Under Policy A, the replications-with-deletion estimate of mean swap-queue wait is 14.49 seconds with a 95% half-width of 0.67 seconds. The batch-means estimate is 14.92 seconds with half-width 0.69 seconds; its chosen batch size is 673, with lag-1 correlation 0.092. The two intervals overlap.

Policy Comparison

Using common random numbers and 30 paired replications, the estimated improvement is 10.37 seconds per drone landing, measured as Policy A minus Policy B. The 95% confidence interval is approximately 9.85 to 10.90 seconds, so zero improvement is not plausible. The paired estimates imply a Policy B mean near 4.12 seconds, compared with 14.49 seconds for Policy A.

Validation and Caveats

Two checks support the analysis: the fleet conservation invariant had maximum violation 0, and the depot Little's Law ratio was 0.9999, essentially 1.0. The main caveat is that this is a steady-state queueing recommendation, so the first 127 minutes of startup behavior are intentionally excluded. The analysis quantifies delay reduction, not purchase price, installation constraints, staffing, or maintenance cost.